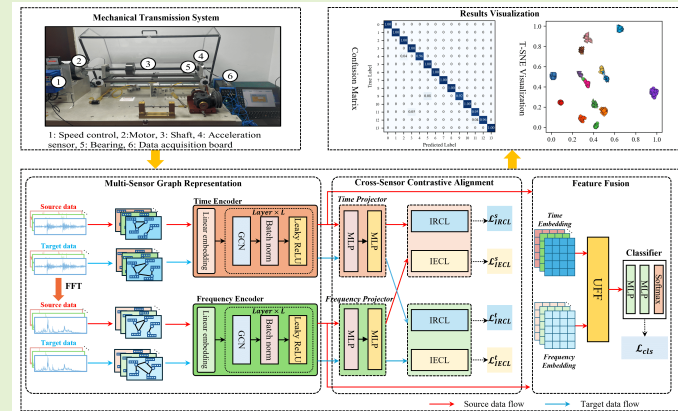


Multi-Sensor Time–Frequency Contrastive Network for Cross-Domain Fault Diagnosis under Limited Labeled Data

Gang Wang, *Member, IEEE*, Zihui Yao, Can Zhang, Xiaohui Yuan, *Senior Member, IEEE*, and Mingfeng Lu, *Student Member, IEEE*

Abstract—Recently, contrastive learning (CL) has been introduced into cross-domain fault diagnosis to learn discriminative representations without extensive labeled data. However, existing methods remain limited in two aspects. First, the heterogeneity and inconsistency among distributed sensors are often overlooked, which hinders the effective exploitation of complementary multi-sensor information. Second, although the inherent relationship between time- and frequency-domain representations provides spectral invariance under varying working conditions, this property has not been fully exploited to alleviate domain shift in cross-domain fault diagnosis. To tackle the aforementioned challenge, a CL-based domain adaptation (DA) framework, named multi-sensor time–frequency contrastive network (MSTF-CN), is proposed to achieve domain-invariant and sensor-consistent feature learning for cross-domain fault diagnosis. Specifically, a multi-sensor graph representation (MSGR) module is designed to capture intrinsic correlations within each sensor by constructing structured graphs for both time-domain signals and frequency-domain representations. Moreover, a cross-sensor contrastive alignment (CSCA) strategy is introduced to explicitly alleviate domain shift and cross-sensor misalignment under varying working conditions. This strategy adopts a dual-alignment mechanism to reduce domain discrepancies and enforce fault representation consistency across sensors. Finally, a unified feature fusion (UFF) module is developed to integrate multi-sensor time–frequency representations, thereby improving feature complementarity and discriminative capability. Extensive experiments conducted on two real-world datasets demonstrate the superiority of the proposed method in multi-sensor cross-domain fault diagnosis tasks.

Index Terms—Multi-sensor fault diagnosis, Domain adaptation, Graph convolutional network, Time-frequency contrast.



I. INTRODUCTION

WITH the rapid development of the Industrial Internet of Things and Cyber-Physical Systems, modern industrial systems are evolving towards high integration and intelligence, making system reliability a critical concern [1], [2]. As the core components of these systems, mechanical equipment is highly susceptible to failures under complex and harsh operating conditions, and such faults directly undermine system reliability, causing unexpected downtime, production losses,

and safety risks [3], [4]. Consequently, the development of accurate and timely fault diagnosis techniques has attracted significant attention from both academia and industry.

Recently, deep learning methods such as convolutional neural networks (CNNs) [5], recurrent neural networks (RNNs) [6], and Transformer-based [7] networks have demonstrated an exceptional capability to automatically learn fault-relevant features from raw sensor signals, enabling effective diagnosis of critical mechanical components. However, these methods are primarily designed for single-sensor data, which fundamentally limits their ability to capture the comprehensive information. In practical industrial systems, such as rotating machinery and gearbox-bearing assemblies, different sensors are often deployed at multiple locations to monitor heterogeneous signals. Meanwhile, environmental disturbances and complex internal mechanisms further increase the difficulty of fault diagnosis, as a single signal often fails to capture the full range of failure modes and distinctive characteristics.

To address these limitations, multi-sensor fault diagnosis has

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Gang Wang, Zihui Yao, Can Zhang, and Mingfeng Lu are all with the School of Management, Hefei University of Technology, Hefei 230009, China (e-mail: wgedison@gmail.com).

Xiaohui Yuan is currently an Associate Professor and Director of Computer Vision and Intelligent Systems Lab, University of North Texas, Denton, TX, USA.

attracted increasing attention, as the integration of multiple sensor signals enables a more comprehensive understanding of equipment conditions and improves diagnostic reliability and accuracy [8]. In particular, graph neural networks (GNNs) [9] have emerged as an effective framework for modeling structured relationships within multi-sensor data, capturing complex interdependencies among sensors and enhancing fault-related representation learning. For example, Yang et al. [10] proposed a GNN-based homogenization framework that leverages GNN to model heterogeneous multi-sensor data, enabling high-quality graph construction and improved machine fault diagnosis performance. Xiao et al. [11] proposed a multi-level information fusion method based on multilayer graph construction and attention-enhanced feature fusion, effectively improving the accuracy of multi-sensor fault diagnosis. Although GNN-based methods have made notable progress in multi-sensor information fusion for fault diagnosis, their application scenarios are generally limited to steady operating conditions, where the source and target data are assumed to follow the same distribution. However, in practical industrial environments, equipment often operates under varying working conditions, resulting in domain shift between the source- and target-domain data.

To mitigate domain shift caused by variable working conditions, transfer learning (TL) has become a research focus of cross-domain fault diagnosis. In particular, DA [12] has been widely adopted, as it leverages labeled data from the source-domain to learn transferable representations that remain effective when deployed on unlabeled data from the target-domain. Current mainstream DA approaches can generally be categorized into statistical-based and adversarial-based methods. Statistical-based methods aim to reduce domain discrepancies using metrics such as maximum mean discrepancy (MMD) [13], correlation alignment (CORAL) [14], and central moment discrepancy (CMD) [15]. For example, Cao et al. [16] proposed a deep domain-adaptive multi-task model that integrates a soft joint MMD metric to jointly reduce marginal and conditional distribution shifts. Wang et al. [17] proposed a deep feature correlation matching network that extends CORAL by aligning first- and second-order feature correlations to improve feature consistency in domains. Si et al. [18] introduced an unsupervised deep transfer learning framework based on CMD, which aligns higher-order statistical characteristics between the source and target domains using time–frequency images. Alternatively, adversarial-based methods introduce a domain discriminator and learn domain-invariant features through an adversarial training process. For example, Zhao et al. [19] developed a joint distribution adaptation network integrated with adversarial learning to simultaneously align marginal and conditional feature distributions between domains. Deng et al. [20] proposed a regularized conditional adversarial adaptation network incorporating feature correlation constraints into the adversarial alignment process, thus improving the domain adaptation capacity under multiple target operating conditions. However, the majority of these approaches primarily emphasize the alignment of the marginal distributions of the source and target domains, while neglecting fine-grained alignment. Even when class-

level domain adaptation is considered, it still cannot guaranty sufficient intra-class compactness and inter-class separability within the feature space, leaving target samples near the decision boundary vulnerable to misclassification.

Contrastive learning [21] has emerged as an effective self-supervised approach to learning discriminative representations without extensive labeled data to remedy the issue of weak intra-class compactness and inter-class separability. By constructing positive and negative sample pairs, CL encourages similar representations for positive pairs while separating negative ones, leading to a more discriminative feature space. Recently, CL has been applied to mechanical fault diagnosis to improve feature discrimination and domain adaptation. For example, Zhou et al. [22] proposed a novel semi-supervised fault diagnosis method using contrastive learning enhanced with multiscale permutation entropy to effectively extract deep features from limited labeled and abundant unlabeled samples. An et al. [23] developed a contrastive domain adaptation method that utilized a contrastive estimation term to reduce the misclassification of samples near the classification boundaries. Moreover, researchers have constructed positive and negative sample pairs from time- and frequency-domain representations, where integrating information from both domains enables the extraction of richer signal features that are not readily available solely in the time domain [24]. For example, Zhang et al. [25] proposed a time–frequency contrast domain adaptation method that leverages LSTM–CNN feature extraction and MMD-based domain alignment to improve train bearing fault diagnosis under Doppler-induced distortion. Wang et al. [26] introduced a frequency-assisted contrastive learning framework that leverages complementary time–frequency views and pseudo-labeled data to learn more discriminative representations and achieve superior fault diagnosis performance under limited labeled samples. However, existing CL-based methods rarely consider the joint effect of domain shift and cross-sensor misalignment, resulting in sufficient feature fusion and limited performance in complex industrial scenarios.

To this end, a DA framework termed MSTF-CN is proposed to learn domain-invariant and sensor-consistent representations by the alignment of multi-sensor time- and frequency-domain information under varying working conditions. First, a MSGR module is designed to model intrinsic correlations within the time-frequency representations of each sensor by constructing structured graphs from both time-domain signals and frequency-domain representations. This enables the framework to effectively preserve the relational structure underlying multi-sensor vibration signals. Second, a CSCA strategy is introduced to jointly improve alignment across domains and sensors. By enforcing intra-sensor contrastive learning (IRCL) to capture the physical consistency between time- and frequency-domain views, and inter-sensor contrastive learning (IECL) to reduce cross-sensor discrepancies, the model is encouraged to learn domain-invariant and sensor-consistent features. Finally, a UFF module is developed to fully integrate complementary information from multi-sensor time–frequency representations, thereby promoting comprehensive and discriminative feature learning for reliable cross-domain fault classification. The main contributions of this article are summarized as follows:

- 1) A novel DA fault diagnosis framework, termed MSTF-CN, is proposed for fault diagnosis under varying working conditions. In this framework, cross-domain alignment of multi-sensor time–frequency features is explicitly exploited to learn domain-invariant and sensor-consistent representations.
- 2) A CSCA strategy is introduced to handle time-domain distribution shifts and cross-sensor misalignments. Intra-sensor contrast aligns time- and frequency-domain views of each sensor to learn domain-invariant representations, while inter-sensor contrast further aligns features across sensors to integrate complementary information.
- 3) Extensive experiments on two real-world datasets verify the superiority of MSTF-CN, demonstrating substantial improvements over state-of-the-art (SOTA) methods in cross-domain fault diagnosis.

The rest of this article is organized as follows. Section II elaborates on the proposed MSTF-CN. Section III illustrates the details of the experiment setup. Comprehensive experimental results and discussions are presented and analyzed in Section IV. Finally, Section V introduces the conclusions and future directions of this work.

II. PROPOSED METHOD

A. Problem Formulation

In this study, raw multi-sensor signals are employed to investigate cross-domain fault diagnosis under varying working conditions, where the primary challenges arise from domain shift and cross-sensor misalignment. Let $D_s = \{(x_{i,s}^{m,T}, x_{i,s}^{m,F}, y_i)\}_{i=1, m=1}^{N_s, M}$ represent the source-domain dataset, where $x_{i,s}^{m,T}$ represents the time-domain raw signal collected by the m -th sensor, $x_{i,s}^{m,F}$ denotes the corresponding frequency-domain feature obtained by the Fast Fourier Transform (FFT), $y_i \in \{1, \dots, C\}$ is the associated fault label and C denotes the total number of health states and N_s is the number of source-domain samples. Unlike the source domain, the target domain is composed of a large-scale unlabeled dataset and a small subset of labeled samples. Let $D_t = \{(x_{i,t}^{m,T}, x_{i,t}^{m,F})\}_{i=1, m=1}^{N_t, M}$ represent unlabeled target-domain dataset, and let $D_{\tilde{t}} = \{(x_{i,\tilde{t}}^{m,T}, x_{i,\tilde{t}}^{m,F}, y_{i,\tilde{t}})\}_{i=1, m=1}^{N_{\tilde{t}}, M}$ represent labeled target-domain dataset, where N_t is the number of unlabeled samples, $N_{\tilde{t}}$ is the number of labeled samples, and $N_{\tilde{t}} \ll N_t$. The learning process is conducted in two stages. In the pre-training stage, the fully labeled source-domain dataset D_s and the large-scale unlabeled target-domain dataset D_t are jointly utilized to learn domain-invariant and sensor-consistent representations. In the fine-tuning stage, only the classifier is fine-tuned using the limited labeled target-domain dataset $D_{\tilde{t}}$, enabling the decision boundary to adapt to target-domain characteristics.

B. Framework

The overall framework of the proposed MSTF-CN, as illustrated in Fig. 1, is designed for multi-sensor cross-domain fault diagnosis under varying working conditions. Its objective

is to learn domain-invariant and sensor-consistent representations, thereby facilitating improved adaptation in cross-domain diagnostic tasks that alleviate the adverse effects of domain shift and cross-sensor misalignment. The proposed framework consists of three main components. First, the MSGR module focuses on modeling the temporal dynamics and sensor-specific structural characteristics of multi-sensor signals. For each sensor, the raw time-domain signals and their corresponding frequency-domain features are independently transformed into graph representations, as directly fusing signals from different sensors may introduce distortion due to their inherent heterogeneity. Subsequently, two GCN-based encoders are employed to process the time- and frequency-domain graph representations. These encoders perform neighborhood aggregation and learn structure-aware latent representations for each sensor and each domain. This dual-branch encoding strategy preserves discriminative information embedded in local temporal–spectral structures without interfering with the distinct properties of time- and frequency-domain representations. Second, CSCA is developed to perform fine-grained multi-sensor cross-domain feature alignment through dual-alignment mechanism, including IRCL and IECL. IRCL enforces alignment between time- and frequency-domain views of the same sensor, which promotes domain-invariant representations by leveraging their physical consistency. Meanwhile, IECL aligns representations across different sensors, reducing sensor-specific biases and facilitating the integration of complementary information. Finally, a UFF module is employed to aggregate the aligned multi-sensor domain-invariant representations. Instead of simple feature concatenation, an attention-based fusion mechanism is adopted to adaptively weight multi-sensor features from both time and frequency domains. By explicitly addressing domain shift and sensor misalignment within a unified framework, MSTF-CN enables effective multi-sensor feature fusion for cross-domain fault diagnosis.

C. Multi-Sensor Graph Representation

In multi-sensor fault diagnosis, signals collected from different sensors often exhibit misalignment due to asynchronous sampling mechanisms and sensor-specific acquisition characteristics [27]. Directly enforcing dimensional alignment through resampling or interpolation may distort intrinsic temporal patterns and compromise fault-related information. Moreover, variations in operating conditions lead to domain shifts between source and target data, which requires a representation capable of preserving sensor-specific characteristics while supporting effective adaptation across domains. To address these challenges, we design a multi-sensor graph construction strategy that performs graph construction independently for each sensor and each domain, enabling the preservation of sensor-dependent characteristics while avoiding cross-sensor interference. For labeled source-domain data, each sample is treated as a graph node. The adjacency matrix of the m -th sensor is defined as:

$$A_s^{m,T}(i, j) = A_s^{m,F}(i, j) = \begin{cases} 1, & y_i = y_j \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

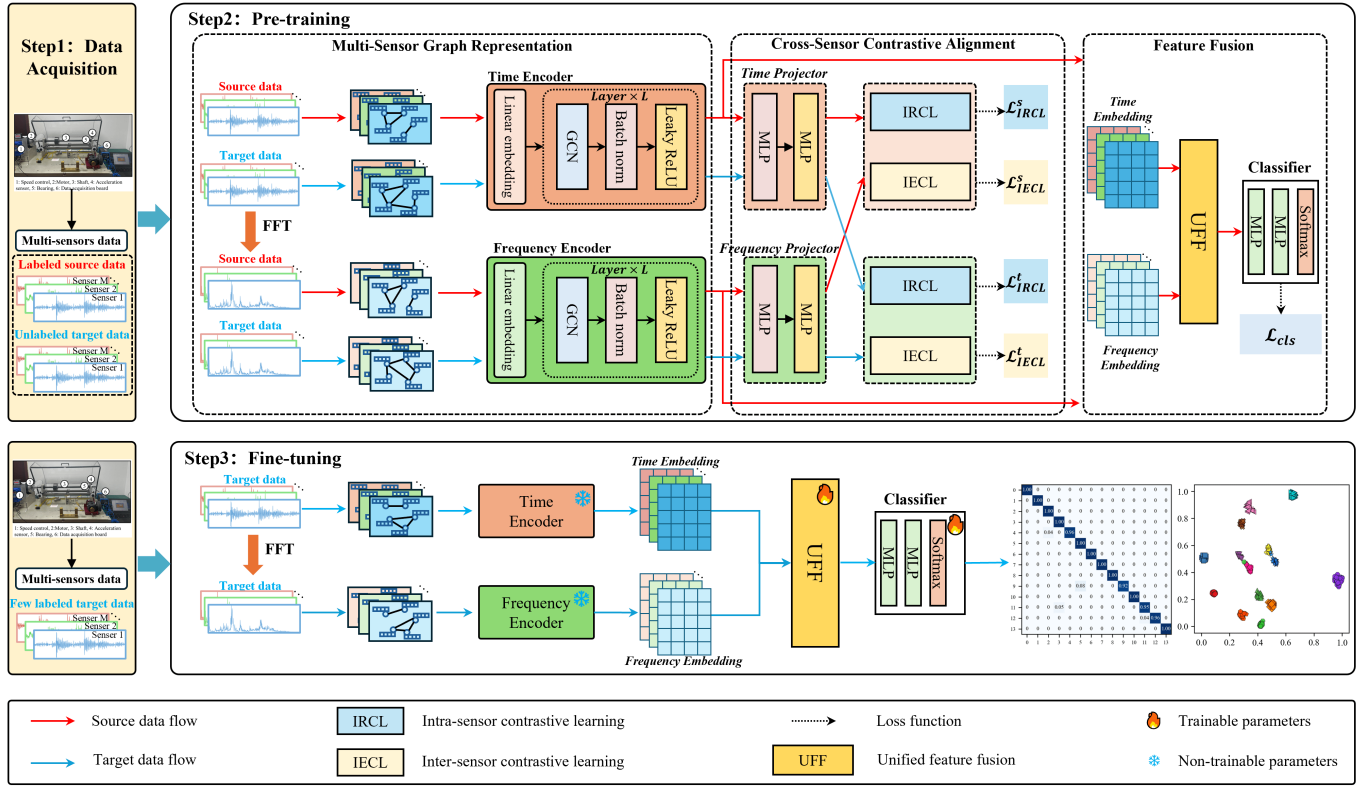


Fig. 1. Overall framework of the proposed MSTF-CN.

This construction explicitly encodes class-level structural relationships within the source domain and facilitates discriminative feature learning under label supervision. For the unlabeled target domain, structural relationships are inferred without label information. Given two time-domain samples $x_{i,t}^{m,T}$ and $x_{j,t}^{m,T}$, their similarity is computed using cosine similarity:

$$\text{Sim} \left(x_{i,t}^{m,T}, x_{j,t}^{m,T} \right) = \frac{x_{i,t}^{m,T} \cdot x_{j,t}^{m,T}}{\|x_{i,t}^{m,T}\|_2 \|x_{j,t}^{m,T}\|_2} \quad (2)$$

Based on the resulting similarity matrix, a k-nearest neighbor strategy is employed to construct local neighborhoods for each node. Specifically, for node i of the m -th sensor in the time domain, its neighborhood is defined as:

$$\mathcal{N}_k^{m,T}(i) = \text{Top-k} \left(\text{Sim} \left(x_{i,t}^{m,T}, x_{j,t}^{m,T} \right) \right) \quad (3)$$

where the $\text{Top-k}(\cdot)$ operator selects the k most similar samples. For node i of the m -th sensor in the frequency domain, its neighborhood is defined as $\mathcal{N}_k^{m,F}(i)$ in the same manner. This strategy constrains the neighborhood size and focuses the graph structure on local relationships, particularly in the absence of label supervision. Accordingly, the adjacency matrix for the target-domain time-domain graph is constructed as:

$$A_t^{m,T}(i,j) = \begin{cases} 1, & j \in \mathcal{N}_k^{m,T}(i) \text{ and } i \in \mathcal{N}_k^{m,T}(j) \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Following the same procedure, the adjacency matrix for target-

domain frequency-domain graph is constructed as:

$$A_t^{m,F}(i,j) = \begin{cases} 1, & j \in \mathcal{N}_k^{m,F}(i) \text{ and } i \in \mathcal{N}_k^{m,F}(j) \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

For each sensor, the time- and frequency-domain graphs are encoded using two encoders, denoted $\mathcal{E}_T(\cdot)$ and $\mathcal{E}_F(\cdot)$, respectively. Both encoders share the same architecture and a weight-sharing strategy is adopted between the source and target domains: all time-domain graphs from the source and target domains are processed by $\mathcal{E}_T(\cdot)$, while all frequency-domain graphs share $\mathcal{E}_F(\cdot)$. This design ensures consistent feature extraction across domains and facilitates knowledge transfer. Each encoder begins with a linear embedding layer that projects raw node features into a unified dimension to accommodate variations in sampling rates, segmentation lengths, and preprocessing across different sensors. For instance, the embedding for time-domain source data from the m -th sensor defined as:

$$H_s^{m,T} = X_s^{m,T} W_s^{m,T} + b_s^{m,T} \quad (6)$$

Following the linear embedding layer, the encoder stacks L GCN blocks. Each block consists of graph convolution, batch normalization, and LeakyReLU activation function. The propagation rule for the l -th layer is expressed as:

$$H_s^{m,T,(l)} = \text{LeakyReLU} \left(\text{BatchNorm} \left(\hat{A}_s^{m,T} H_s^{m,T,(l-1)} W^{T,(l)} \right) \right) \quad (7)$$

where $\hat{A}_s^{m,T} = \left(\tilde{D}_s^{m,T} \right)^{-\frac{1}{2}} \left(A_s^{m,T} + I \right) \left(\tilde{D}_s^{m,T} \right)^{-\frac{1}{2}}$ denotes

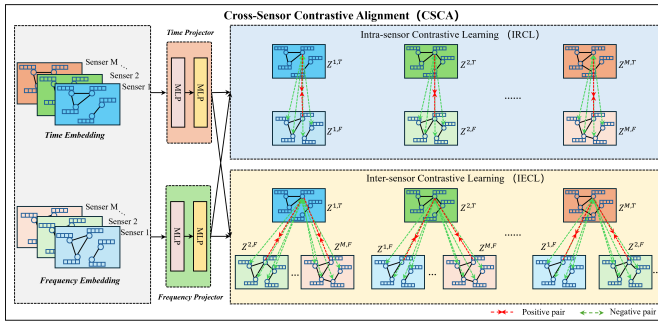


Fig. 2. Working mechanism of the proposed CSCA framework.

the normalized adjacency matrix, I is the identity matrix for self-loops, $\hat{D}_s^{m,T}$ is the degree matrix, and $W^{T,(l)}$ is the learnable weight matrix for the l -th layer of time encoder. The frequency-domain encoder $\mathcal{E}_F(\cdot)$ follows the same computational structure. After L layers, the encoders output node embeddings that encode sensor-wise spectral characteristics and intra-sensor dependency relationships embedded in the multi-sensor measurements. For clarity, the final outputs are summarized as follows:

$$H_s^{m,T} = \mathcal{E}_T(X_s^{m,T}, A_s^{m,T}), \quad H_s^{m,F} = \mathcal{E}_F(X_s^{m,F}, A_s^{m,F}) \quad (8)$$

$$H_t^{m,T} = \mathcal{E}_T(X_t^{m,T}, A_t^{m,T}), \quad H_t^{m,F} = \mathcal{E}_F(X_t^{m,F}, A_t^{m,F}) \quad (9)$$

D. Cross-Sensor Contrastive Alignment

In cross-domain multi-sensor fault diagnosis, representations learned from different sensors and different signal domains are often misaligned due to operating-condition variations and heterogeneous sensing characteristics. Although these representations correspond to the same underlying fault state, their feature distributions can differ substantially, weakening the consistency of fault semantics and degrading diagnostic performance. Therefore, explicitly aligning multi-sensor representations is essential for learning domain-invariant and sensor-consistent features. Contrastive learning provides an effective mechanism for such alignment by directly modeling pairwise relationships between representations [21]. By constructing appropriate positive and negative pairs, contrastive objectives encourage representations conveying the same fault semantics to be close in the embedding space while separating those associated with different fault states. In this work, contrastive learning is explicitly designed as a mechanism for structured alignment across both domains and sensors, enabling the learned representations to simultaneously satisfy domain invariance and sensor consistency, and the specific working mechanism is shown in Fig. 2.

To support stable and effective contrastive optimization, a projection head is further introduced as an auxiliary component. This design decouples representation learning for fault diagnosis from contrastive optimization, which has been shown to improve training stability and convergence in contrastive frameworks. Specifically, the projection function $Proj(\cdot)$ is implemented as a lightweight multilayer perceptron

consisting of linear layers, batch normalization, and nonlinear activations. Following common practice, the projection head is used only during contrastive training and discarded during inference, where the encoder output is directly fed into the classifier for fault diagnosis. Formally, the projected time- and frequency-domain representations for the source and target domains are given by

$$Z_s^{m,T} = Proj_T(H_s^{m,T}), \quad Z_s^{m,F} = Proj_F(H_s^{m,F}) \quad (10)$$

$$Z_t^{m,T} = Proj_T(H_t^{m,T}), \quad Z_t^{m,F} = Proj_F(H_t^{m,F}) \quad (11)$$

Based on these projected representations, two contrastive learning objectives are designed, namely intra-sensor contrastive learning and inter-sensor contrastive learning, which respectively address domain shift and cross-sensor misalignment.

1) Intra-sensor Contrastive Learning: Under varying operating conditions, fault-related representations in time-domain signals are often distorted. In contrast, according to signal analysis theory, the relationship between time domain and frequency domain provides an invariance that remains effective under varying operating conditions [28]. Motivated by this observation, IRCL exploits the inherent consistency between time- and frequency-domain representations to enhance the learning of transferable sensor-specific features. Specifically, IRCL enforces alignment between the time- and frequency-domain representations extracted from the same sensor, encouraging the model to capture consistent fault-related characteristics shared between the two domains.

In IRCL, positive and negative pairs are constructed from the projected time- and frequency-domain representations in the same sensor. This contrastive formulation encourages representations to cluster closely, while preserving discriminative separation between different fault states. As a result, domain shift caused by operating-condition variations is effectively alleviated through domain-level alignment. For labeled source-domain samples, supervised IRCL exploits class labels to enhance fault semantic consistency across domains. The supervised IRCL loss is defined as:

$$\mathcal{L}_{IRCL}^s = -\frac{1}{MN_s} \sum_{m=1}^M \sum_{i=1}^{N_s} \frac{1}{|P(i)|} \sum_{p \in P(i)} \frac{\exp(z_{i,s}^{m,T} \cdot z_{p,s}^{m,F} / \tau)}{\sum_{a \in A(i)} \exp(z_{i,s}^{m,T} \cdot z_{a,s}^{m,F} / \tau)} \quad (12)$$

where $P(i)$ denotes the set of source-domain samples that share the same fault label as the i -th sample, $|P(i)|$ denotes the number of valid positive samples, $A(i)$ denotes the set of all candidate samples excluding the anchor, and τ is the temperature parameter that controls the sharpness of similarity distribution.

For unlabeled target-domain samples, IRCL is formulated in an unsupervised manner by constructing instance-level positive pairs from time- and frequency-domain representations of the

same sample. The unsupervised IRCL loss is given by:

$$\mathcal{L}_{\text{IRCL}}^t = -\frac{1}{MN_t} \sum_{m=1}^M \sum_{i=1}^{N_t} \log \frac{\exp(z_{i,t}^{m,T} \cdot z_{i,t}^{m,F} / \tau)}{\sum_{a=1}^{N_t} \mathbf{1}_{[a \neq i]} \exp(z_{i,t}^{m,T} \cdot z_{a,t}^{m,F} / \tau)} \quad (13)$$

Through joint supervised and unsupervised optimization, IRCL aligns time- and frequency-domain representations between source and target domains, thereby reducing domain shift and promoting domain-invariant representations under varying operating conditions.

2) Inter-sensor Contrastive Learning: Although multiple sensors monitor the same underlying health condition, their observations differ due to distinct sensing mechanisms and response characteristics. As a result, the extracted features may reside in heterogeneous semantic spaces, necessitating explicit sensor-level alignment. To address this issue, IECL is introduced to explicitly enforce sensor-level consistency. Specifically, IECL aligns representations corresponding to the same fault state across different sensors by pulling them closer in the embedding space, while maintaining separation between representations associated with different fault states. Through this cross sensor contrastive constraint, sensor-induced variations are suppressed and fault-related representations that are shared across sensors are emphasized. As a result, multi-sensor observations are mapped into a unified and sensor-consistent representation space, enabling different sensors to contribute coherently to subsequent learning stages.

In IECL, positive and negative pairs are constructed from the projected time- and frequency-domain representations across different sensors. This cross sensor contrastive formulation encourages the alignment of embeddings associated with the same fault class between sensors. For labeled source-domain samples, the supervised IECL loss is expressed as:

$$\mathcal{L}_{\text{IECL}}^s = -\frac{1}{M(M-1)N_s} \sum_{m_1=1}^M \sum_{m_2=1}^M \sum_{i=1}^{N_s} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_{i,s}^{m_1,T} \cdot z_{p,s}^{m_2,F} / \tau)}{\sum_{a \in A(i)} \exp(z_{i,s}^{m_1,T} \cdot z_{a,s}^{m_2,F} / \tau)} \quad (14)$$

where $m_1 \neq m_2$.

For unlabeled target-domain samples, the unsupervised IECL loss is expressed as:

$$\mathcal{L}_{\text{IECL}}^t = -\frac{1}{M(M-1)N_t} \sum_{m_1=1}^M \sum_{m_2=1}^M \sum_{i=1}^{N_t} \log \frac{\exp(z_{i,t}^{m_1,T} \cdot z_{i,t}^{m_2,F} / \tau)}{\sum_{a=1}^{N_t} \mathbf{1}_{[a \neq i]} \exp(z_{i,t}^{m_1,T} \cdot z_{a,t}^{m_2,F} / \tau)} \quad (15)$$

Through these, IECL aligns fault representations across heterogeneous sensors in both source and target domains, thereby promoting sensor-consistent fault semantics despite differences in sensing mechanisms and response characteristics.

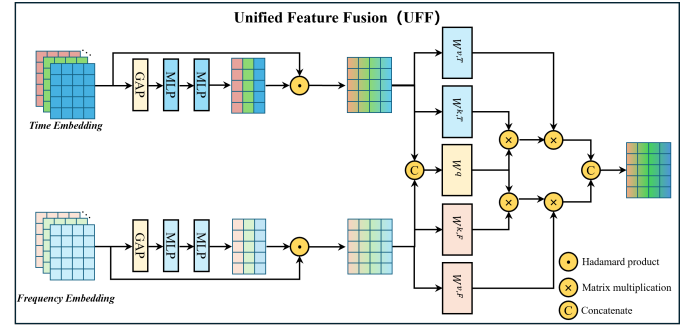


Fig. 3. Structure of the proposed UFF module.

3) Total Contrastive Loss: IRCL enforces time–frequency consistency within each sensor, enabling the extraction of stable fault-related representations that adapt to different operating conditions. IECL complements this process by aligning representations across sensors, thereby reducing discrepancies caused by sensing mechanisms and response characteristics. Together, these two contrastive schemes shape a unified embedding space that captures global discriminative structure while suppressing domain- and sensor-specific variations. By jointly optimizing IRCL and IECL, the proposed contrastive learning strategy establishes a unified embedding space in which fault-related representations are consistently expressed across domains and sensors. The overall objective of contrastive learning is defined as follows:

$$\mathcal{L}_{cl} = \lambda_1 (\mathcal{L}_{\text{IRCL}}^s + \mathcal{L}_{\text{IRCL}}^t) + \lambda_2 (\mathcal{L}_{\text{IECL}}^s + \mathcal{L}_{\text{IECL}}^t) \quad (16)$$

where λ_1 and λ_2 are weighting coefficients that balance the contributions of IRCL and IECL.

E. Feature Fusion

To fully exploit the complementary information embedded across multiple sensors and time–frequency signal domains, the UFF module is designed to adaptively refine multi-sensor and time–frequency features, and the specific structure is shown in Fig. 3. First, the features extracted from each sensor are aggregated via a channel-attention mechanism to adaptively emphasize informative sensors while suppressing less relevant or noisy ones. Formally, given the projected node embeddings of all sensors $\{H^{m,T}\}_{m=1}^M$, the channel attention mechanism computes sensor-wise importance weights using global pooling followed by a gating function:

$$\mathbf{w}^T = \sigma(W_2^T \delta(W_1^T \text{GAP}([H^{1,T}, \dots, H^{M,T}])))) \quad (17)$$

$$\mathcal{H}^T = \sum_{m=1}^M w^{m,T} \odot H^{m,T} \quad (18)$$

where $[H^{1,T}, \dots, H^{M,T}]$ denotes the concatenation of sensor embeddings along the channel dimension, $\text{GAP}(\cdot)$ is global average pooling, W_1^T and W_2^T are learnable weight matrices of the gating network, $\delta(\cdot)$ is a ReLU activation, $\sigma(\cdot)$ is the sigmoid function, $\mathbf{w}^T = [w^{1,T}, \dots, w^{M,T}]$ assigns an adaptive importance score to each sensor based on its temporal discriminative capacity, $\odot$ denotes Hadamard product,

and $\mathcal{H}^T$ represents the fused sensor-level feature embedding. Frequency-domain fusion is defined analogously with parameters W_1^T and W_2^T to produce $\mathcal{H}^F$.

$$\mathbf{w}^F = \sigma(W_2^F \delta(W_1^F GAP([H^{1,F}, \dots, H^{M,F}])) \quad (19)$$

$$\mathcal{H}^F = \sum_{m=1}^M w^{m,F} \odot H^{m,F} \quad (20)$$

To further integrate complementary information contained in the temporal and spectral representations, a cross-attention mechanism is employed to enable bidirectional interaction between the fused time-domain feature $\mathcal{H}^T$ and frequency-domain feature $\mathcal{H}^F$. Unlike simple concatenation or averaging, cross-attention explicitly models how temporal features attend to spectrally discriminative components and how spectral features attend to temporally salient structures, thereby enhancing the representation quality in a more fine-grained and adaptive manner. Specifically, for each domain, learnable linear projections are used to generate key and value matrices. For the time-domain branch, the key and value matrices are computed as:

$$K^T = \mathcal{H}^T W^{k,T}, \quad V^T = \mathcal{H}^T W^{v,T} \quad (21)$$

And for the frequency-domain branch, the projections are similarly defined as:

$$K^F = \mathcal{H}^F W^{k,F}, \quad V^F = \mathcal{H}^F W^{v,F} \quad (22)$$

To enable cross-domain interactions, a unified query matrix is generated by concatenating the temporal and spectral features followed by a shared projection:

$$Q = [\mathcal{H}^T, \mathcal{H}^F] W^q \quad (23)$$

where, $W^{k,T}$, $W^{v,T}$, $W^{k,F}$, $W^{v,F}$ and W^q are trainable projection matrices that map the time, frequency, and concatenated representations into a shared latent space with the same embedding dimension.

Based on these, the cross-attention module computes two complementary attention responses. The first response captures how the fused query attends to time-domain information, while the second captures the attention toward frequency-domain information. The cross-attention is computed using the following formula:

$$\mathcal{H}_{UFF}^T = \text{Softmax}\left(\frac{Q(K^T)^\top}{\sqrt{D^T}}\right) V^T \quad (24)$$

$$\mathcal{H}_{UFF}^F = \text{Softmax}\left(\frac{Q(K^F)^\top}{\sqrt{D^F}}\right) V^F \quad (25)$$

where $\mathcal{H}_{UFF}^T$ and $\mathcal{H}_{UFF}^F$ represent the features after cross-attention, $\sqrt{D^T}$ and $\sqrt{D^F}$ is a scaling factor to mitigate the problem of excessively large dot product values. Subsequently, we concatenate the features to achieve a more comprehensive representation, as given by the following formula:

$$S_{UFF} = [\mathcal{H}_{UFF}^T, \mathcal{H}_{UFF}^F] \quad (26)$$

TABLE I

SPECIFIC INFORMATION OF THE PU BEARING DATASET

Condition	Speed (rpm)	Load (Nm)	Force (N)	No. of samples	Fault category
A1	1500	0.7	1000	3505	K001, KA04, KA15, KA16,
A2	900	0.7	1000	3505	KA22, KA30, KB23, KB24,
A3	1500	0.1	1000	3505	KB27, KI04, KI16, KI17,
A4	1500	0.7	400	3505	KI18, KI21

TABLE II

SPECIFIC INFORMATION OF THE HUST BEARING DATASET

Condition	Speed (Hz)	No. of samples	Fault category
B1	20	2304	H, 0.5X_B, 0.5X_C,
B2	30	2304	0.5X_I, 0.5X_O, B,
B3	40	2304	C, I, O
B4	60	2304	

The fused feature S_{UFF} is then fed into a classifier and the loss of cross-entropy classification is calculated to update the fault diagnosis network.

$$\hat{Y} = \text{Classifier}(S_{UFF}) \quad (27)$$

$$\mathcal{L}_{cls} = -\frac{1}{N_s} \sum_{i=1}^{N_s} \sum_{c=1}^C y_{i,c} \log \hat{y}_{i,c} \quad (28)$$

where $\text{Classifier}(\cdot)$ consists of two layers of Multilayer Perceptron (MLP), $\hat{Y} \in R^{N_s \times C}$ denotes the predicted class probability matrix, $\hat{y}_{i,c}$ is the predicted class probability of the i -th sample belonging to class c and $y_{i,c}$ is the one-hot encoded ground-truth label.

F. Training Procedure

To jointly optimize the performance of contrastive learning and classification, the total training objective combines contrast loss $\mathcal{L}_{cl}$ and classification loss $\mathcal{L}_{cls}$. The overall loss function is expressed as:

$$\mathcal{L}_{total} = \lambda \mathcal{L}_{cl} + (1 - \lambda) \mathcal{L}_{cls} \quad (29)$$

where λ are weighting coefficients that balance the contributions of contrastive learning and supervised classification. During the fine-tuning phase, the encoder parameters are frozen to preserve the domain-invariant and sensor-consistent representations learned during pre-training, while the UFF module and classifier are further optimized using limited labeled target data. This design allows the fusion mechanism and the decision boundary to adapt to target-domain characteristics without disrupting the learned representation space.

III. EXPERIMENT SETUP

This section provides an overview of experimental datasets from two different bearing dataset, as well as the evaluation criteria and experimental procedure.

Algorithm 1 Training procedure of MSTF-CN

- 1: **Input:** Labeled source data $D_s = \{(x_{i,s}^{m,T}, x_{i,s}^{m,F}, y_i)\}_{i=1, m=1}^{N_s, M}$; unlabeled target data $D_t = \{(x_{i,t}^{m,T}, x_{i,t}^{m,F})\}_{i=1, m=1}^{N_t, M}$; few labeled target data $D_{\bar{t}} = \{(x_{i,\bar{t}}^{m,T}, x_{i,\bar{t}}^{m,F}, y_{i,\bar{t}})\}_{i=1, m=1}^{N_{\bar{t}}, M}$; number of sensors M ; neighbors k ; temperature τ ; learning rate lr ; epochs $epochs_1, epochs_2$.
- 2: Construct multi-sensor graphs $A_s^{m,T}, A_s^{m,F}, A_t^{m,T}, A_t^{m,F}$ according to (1)-(5).
- 3: **// Pretraining**
- 4: Initialize $\mathcal{E}_T(\cdot), \mathcal{E}_F(\cdot)$, projection heads $\text{Proj}_T(\cdot), \text{Proj}_F(\cdot)$, UFF module and classifier.
- 5: **for** $epoch = 1$ to $epochs_1$ **do**
- 6: Extract representations using $\mathcal{E}_T(\cdot)$ and $\mathcal{E}_F(\cdot)$.
- 7: Project into contrastive space by $\text{Proj}_T(\cdot)$ and $\text{Proj}_F(\cdot)$.
- 8: Compute IRCL and IECL losses according to (12)-(16).
- 9: Fuse features using UFF module according to (17)-(26).
- 10: Compute classification loss on source samples.
- 11: Update parameters by minimizing total loss (29).
- 12: **end for**
- 13: **// Fine-tuning**
- 14: Load pre-trained parameters of $\mathcal{E}_T(\cdot), \mathcal{E}_F(\cdot)$, UFF and classifier.
- 15: Freeze $\mathcal{E}_T(\cdot)$ and $\mathcal{E}_F(\cdot)$.
- 16: **for** $epoch = 1$ to $epochs_2$ **do**
- 17: Extract and fuse target features.
- 18: Update UFF and classifier using cross-entropy loss.
- 19: **end for**
- 20: **// Test**
- 21: Load trained parameters.
- 22: Extract target-domain features using $\mathcal{E}_T(\cdot)$ and $\mathcal{E}_F(\cdot)$.
- 23: Fuse features by UFF module.
- 24: Predict target-domain labels using classifier.
- 25: **Output:** Predicted fault labels for target-domain samples.

A. Description of the dataset

1) **PU Dataset:** The Paderborn University (PU) bearing dataset [29] is widely used for fault diagnosis research due to its realistic fault types and multiple operating conditions. The dataset is collected from a bearing test rig equipped with multiple sensors, including vibration and current sensors, under controlled experimental settings. The test platform consists of a motor, torque shaft, bearing module, flywheel, and load motor, as illustrated in Fig. 4. Four operating conditions, differing in rotating speed, load torque, and radial force, are selected in this study, as summarized in Table I. These variations introduce significant distribution discrepancies, making the dataset suitable for cross-domain fault diagnosis evaluation. A total of 14 categories are considered, including one healthy condition and thirteen real fault categories generated during accelerated degradation. Such diversity introduces substantial intra-class variation and inter-class similarity, posing significant challenges for reliable fault classification.

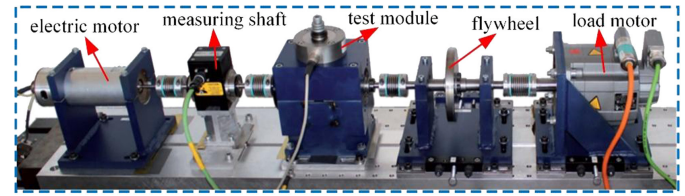


Fig. 4. Experimental platform of the PU dataset.

2) **HUST Dataset:** The Huazhong University of Science and Technology (HUST) bearing dataset [30] was collected using a Spectra-Quest mechanical fault simulator. The experimental platform consists of a motor, shaft, a three-way accelerometer, bearing housing, and a data acquisition system, which enables stable acquisition of three-channel vibration signals under various operating conditions. The experimental setup is illustrated in Fig. 5. In this study, four operating conditions with different rotational speeds are selected from the dataset, corresponding to 20 Hz, 30 Hz, 40 Hz, and 60 Hz, as summarized in Table II. Under these operating conditions, nine fault categories are considered, including one healthy category and eight fault categories with different fault types and severities. Specifically, the selected categories include medium ball fault, medium combination fault, medium inner race fault, medium outer race fault, ball fault, combination fault, inner race fault, and outer race fault. Detailed descriptions of the selected fault categories.

B. Evaluation Criteria

To comprehensively assess fault classification performance, two widely used evaluation metrics are employed: Accuracy and F1-score. Accuracy reflects the overall classification correctness, while F1-score provides a balanced evaluation by jointly considering precision and recall, which is particularly important for multi-class fault diagnosis tasks with potential class imbalance.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (30)$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (31)$$

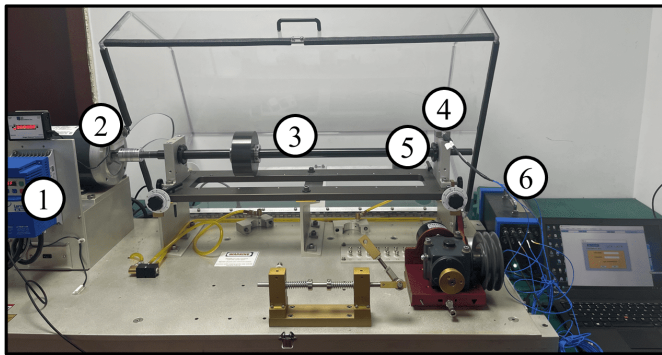
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (32)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (33)$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

C. Experimental Procedure

To evaluate the effectiveness of the proposed method in cross-domain fault diagnosis, experiments are conducted on both the PU and HUST bearing datasets under cross-domain settings. For the PU dataset, four operating conditions denoted as A1–A4 are considered, while for the HUST dataset, four operating conditions denoted as B1–B4 are used. By constructing transfer tasks between every pair of different operating conditions, 12 cross-domain transfer tasks are obtained for each dataset. This protocol enables a systematic evaluation



1: Speed control, 2: Motor, 3: Shaft, 4: Acceleration sensor, 5: Bearing, 6: Data acquisition board

Fig. 5. Experimental platform of the HUST dataset.

of the model's adaptation capability under varying operating conditions. The proposed MSTF-CN is compared with a diverse set of baseline methods representing different learning paradigms for cross-domain fault diagnosis. Specifically, MK-MMD [31], CORAL [14], and DANN [32] are selected as classical domain adaptation baselines. MoCo [33] and SimCLR [34] are included as representative contrastive learning baselines. In addition, WSCN [35], DACN [36], CA-DGNN [37], MCDDFA [38], TF-C [39], and CCLP [40] are adopted as recent deep learning baselines for fault diagnosis.

To ensure a fair and reproducible evaluation, all methods are implemented under the same experimental protocol. Specifically, the dataset is divided into training, validation, and testing sets with a ratio of 8:1:1. For methods adopting a pre-training and fine-tuning paradigm, including the proposed method, one-eighth of the training set in target-domain is randomly selected as labeled target-domain samples for fine-tuning. The same target-label budget is adopted for all applicable comparison methods to ensure a fair comparison. For the proposed framework, graph construction is performed using all samples within each domain. During model optimization, only the training nodes participate in forward propagation and loss computation according to the predefined training mask.

The model is optimized using the Adam optimizer with a learning rate of 0.001 and a weight decay of 0.0001. The maximum number of training epochs is set to 100. Early stopping is employed to alleviate overfitting, where the training process is terminated if the validation accuracy does not improve for 10 consecutive epochs, and the model with the best validation performance is selected for final testing. To reduce the influence of randomness, each experiment is repeated ten times with different random seeds, and the average classification accuracy together with the corresponding standard deviation is reported. The key hyperparameters of the proposed method are determined using a conservative tuning strategy. Specifically, grid search is conducted for the number of nearest neighbors k for graph construction and the loss weight λ balancing the contrastive loss and the classification loss, resulting in $k = 10$ and $\lambda = 0.6$. The weighting coefficients of the intra-sensor and inter-sensor contrastive losses are fixed at $\lambda_1 = \lambda_2 = 0.5$. The temperature parameter in the contrastive

loss is fixed to the commonly adopted value of $\tau = 0.08$. For all comparison methods, the hyperparameter settings follow those recommended in the original papers.

IV. RESULTS AND DISCUSSIONS

A. Results

Table III and Table IV present the accuracy for the various methods evaluated on the PU and HUST datasets, respectively. The best results are highlighted in bold, and the second-best results are underlined. It can first be observed that the proposed MSTF-CN achieves the best overall performance across all transfer tasks on both datasets. Taking the average ACC as an example, MSTF-CN improves over the suboptimal method by 1.76% on the PU dataset and 0.78% on the HUST dataset, respectively, demonstrating its superior cross-domain fault diagnosis capability. The comparison results indicate that learning discriminative representations through contrastive learning is generally more effective than relying solely on conventional distribution alignment. However, directly applying generic contrastive learning strategies still yields limited improvements because the positive and negative sample pairs are constructed without considering the characteristics of multi-sensor fault signals. Consequently, the learned representations are insufficient to capture complementary information across sensors and domains. In contrast, the proposed MSTF-CN jointly exploits complementary time- and frequency-domain information and performs both intra-sensor and inter-sensor contrastive alignment. By designing contrastive objectives according to the characteristics of multi-sensor fault diagnosis, the proposed method effectively encourages representations with consistent fault semantics while preserving complementary information among different sensors. Meanwhile, the graph representation further captures structural relationships among samples, enabling more robust feature learning under domain shifts. These advantages allow MSTF-CN to consistently achieve superior performance across different transfer tasks and datasets.

B. Discussions

1) *Visualization Analysis*: To provide a qualitative evaluation of classification performance under cross-domain transfer, confusion matrices of selected representative methods are presented in Fig. 6–7 for the PU and HUST datasets, respectively. The visualizations are presented for a representative transfer task ($A1 \rightarrow A3$ and $B1 \rightarrow B3$), which is selected due to its relatively stable across all compared methods, enabling clearer comparison of classification performance. In each confusion matrix, the vertical axis represents the ground-truth fault labels, while the horizontal axis denotes the predicted labels, and the diagonal elements correspond to correctly classified ratio for each fault category. As shown in Fig. 6-7, the confusion matrices produced by the proposed MSTF-CN exhibit a clear and concentrated diagonal structure, where most samples are correctly classified. This demonstrates that MSTF-CN is capable of learning highly discriminative and domain-invariant representations under varying operating conditions. In contrast, baseline methods exhibit noticeable misclassifications among several fault categories, particularly for classes with similar

TABLE III
EXPERIMENTAL RESULTS OF THE CROSS-DOMAIN TASK ON THE PU DATASET

Task	A1→A2	A1→A3	A1→A4	A2→A1	A2→A3	A2→A4	A3→A1	A3→A2	A3→A4	A4→A1	A4→A2	A4→A3	Avg
MK-MMD	28.58±2.01	69.29±1.99	56.32±3.53	40.17±2.75	42.94±2.18	39.09±2.07	70.86±3.10	26.65±3.32	57.81±1.92	50.31±1.63	34.26±2.14	49.77±1.95	47.17
CORAL	33.15±2.21	91.75±0.33	68.58±1.43	48.20±1.01	44.69±1.02	39.86±1.71	90.03±0.37	30.80±2.12	67.92±1.45	65.20±1.37	34.83±2.79	61.50±1.21	56.37
DANN	36.79±2.94	93.61±0.68	74.53±1.69	64.94±1.58	64.77±2.09	52.68±1.80	94.71±0.34	33.75±1.46	79.72±2.14	71.66±1.68	39.86±2.10	70.71±1.60	64.81
MoCo	79.60±1.98	84.89±2.72	85.01±1.88	88.91±1.92	83.84±2.58	84.87±1.80	87.54±2.24	78.27±2.17	84.53±2.42	88.37±2.66	79.38±2.52	86.05±2.02	84.27
SimCLR	77.84±2.70	84.41±2.65	84.73±1.19	87.17±2.55	85.23±1.81	82.82±1.77	88.11±2.49	79.63±2.03	83.48±1.77	88.29±1.71	78.32±1.88	85.88±1.93	83.82
WSCN	70.75±2.12	94.59±1.51	86.73±1.34	85.77±1.30	89.83±1.76	75.33±1.77	92.66±1.26	78.86±2.03	87.69±1.64	86.23±0.95	78.62±1.18	84.01±1.44	84.25
DACN	73.98±2.20	95.48±1.01	89.05±1.29	89.68±1.21	90.40±1.02	75.13±1.44	91.60±0.53	75.23±2.13	85.23±2.08	91.25±1.41	80.51±1.18	85.68±1.69	85.27
CA-DGNN	65.26±2.36	95.34±0.60	92.91±2.85	90.89±1.47	91.33±1.39	72.88±2.18	94.60±0.50	78.10±2.07	87.31±2.79	88.00±1.86	77.97±2.21	89.86±1.99	85.37
MCDDFA	64.43±2.44	98.03±0.42	93.28±2.45	89.60±1.42	93.31±0.72	76.17±1.91	96.99±0.27	77.76±2.31	88.38±2.75	88.74±2.34	79.29±1.64	85.99±2.13	85.99
TF-C	83.04±2.69	96.92±0.76	94.87±0.72	93.97±1.18	94.41±1.06	93.44±1.08	93.77±0.89	82.78±1.97	94.90±0.50	93.06±0.85	81.21±1.76	94.32±0.88	91.39
CCLP	85.16±1.92	97.56±1.15	95.84±1.25	91.20±1.86	93.59±1.19	91.48±1.13	93.94±1.67	84.03±1.43	<u>95.01±0.89</u>	<u>94.40±0.94</u>	<u>82.10±1.28</u>	<u>96.16±1.45</u>	<u>91.70</u>
MSTF-CN	86.34±2.11	98.10±1.03	<u>95.34±1.59</u>	94.19±1.09	97.37±0.71	94.75±1.58	<u>94.64±1.13</u>	87.36±1.97	95.37±0.81	94.67±1.45	85.68±0.96	97.82±0.54	93.46

TABLE IV
EXPERIMENTAL RESULTS OF THE CROSS-DOMAIN TASK ON THE HUST DATASET

Task	B1→B2	B1→B3	B1→B4	B2→B1	B2→B3	B2→B4	B3→B1	B3→B2	B3→B4	B4→B1	B4→B2	B4→B3	Avg
MK-MMD	95.58±1.39	68.23±2.65	55.80±3.01	51.95±3.06	84.85±2.78	82.38±2.49	53.98±3.23	85.89±2.47	90.35±1.16	38.18±3.97	52.64±4.10	71.65±2.85	69.29
CORAL	95.15±1.33	65.28±2.38	66.71±2.72	74.55±1.87	91.08±0.93	81.21±2.82	51.13±4.01	89.17±2.10	92.47±1.73	39.74±4.69	55.99±4.07	82.03±1.61	73.70
DANN	93.07±1.10	73.16±2.30	69.52±2.65	72.60±2.99	94.76±0.94	93.59±2.75	64.20±3.30	80.09±2.19	93.51±0.98	45.76±2.89	66.34±2.97	82.99±2.15	77.46
MoCo	96.92±1.28	93.16±2.45	90.82±2.38	91.08±2.74	92.99±1.35	91.69±1.48	92.42±1.67	97.66±1.38	91.17±1.26	91.30±1.58	98.53±0.48	92.90±1.00	93.38
SimCLR	97.23±1.12	92.21±2.09	89.91±2.62	92.21±2.66	93.07±1.54	88.10±1.30	93.33±2.19	96.23±1.12	93.38±1.25	90.69±1.28	98.18±1.32	93.51±1.95	93.17
WSCN	97.65±1.27	94.11±1.86	89.00±1.56	87.97±2.07	89.83±2.11	87.40±2.43	91.77±1.77	96.02±2.89	95.93±1.64	85.76±1.78	89.57±1.40	91.09±1.85	91.34
DACN	97.60±1.48	97.14±0.80	85.23±1.79	85.13±2.02	90.40±1.35	93.91±1.33	92.25±1.90	95.23±1.57	97.6±0.92	83.86±1.83	88.46±1.90	90.68±1.31	91.46
CA-DGNN	97.74±1.81	93.92±1.00	89.15±1.87	87.71±2.55	95.17±2.23	97.51±1.41	90.59±3.50	98.52±1.12	98.39±1.69	83.07±2.29	88.57±1.28	89.06±0.96	92.45
MCDDFA	99.57±0.39	95.50±1.63	90.58±1.98	93.23±1.81	97.71±1.61	98.10±1.50	91.69±2.16	98.87±0.44	<u>99.57±0.99</u>	81.36±2.89	89.78±1.91	91.04±1.03	93.91
TF-C	97.66±1.23	97.40±1.00	99.65±0.47	98.10±1.10	99.74±0.29	99.61±0.49	95.13±0.94	97.84±1.71	<u>99.44±0.64</u>	95.69±1.52	97.27±1.11	98.61±0.45	98.01
CCLP	98.18±1.22	99.44±0.57	98.06±1.41	99.30±0.42	98.09±1.33	99.46±0.63	99.57±0.34	99.52±0.53	99.16±0.36	97.61±1.49	98.44±0.79	98.66±0.92	98.78
MSTF-CN	99.78±0.34	99.57±0.43	100±0.00	99.48±0.26	99.78±0.22	100±0.00	99.05±0.60	99.74±0.35	100±0.00	97.75±0.50	99.83±0.29	99.78±0.29	99.56

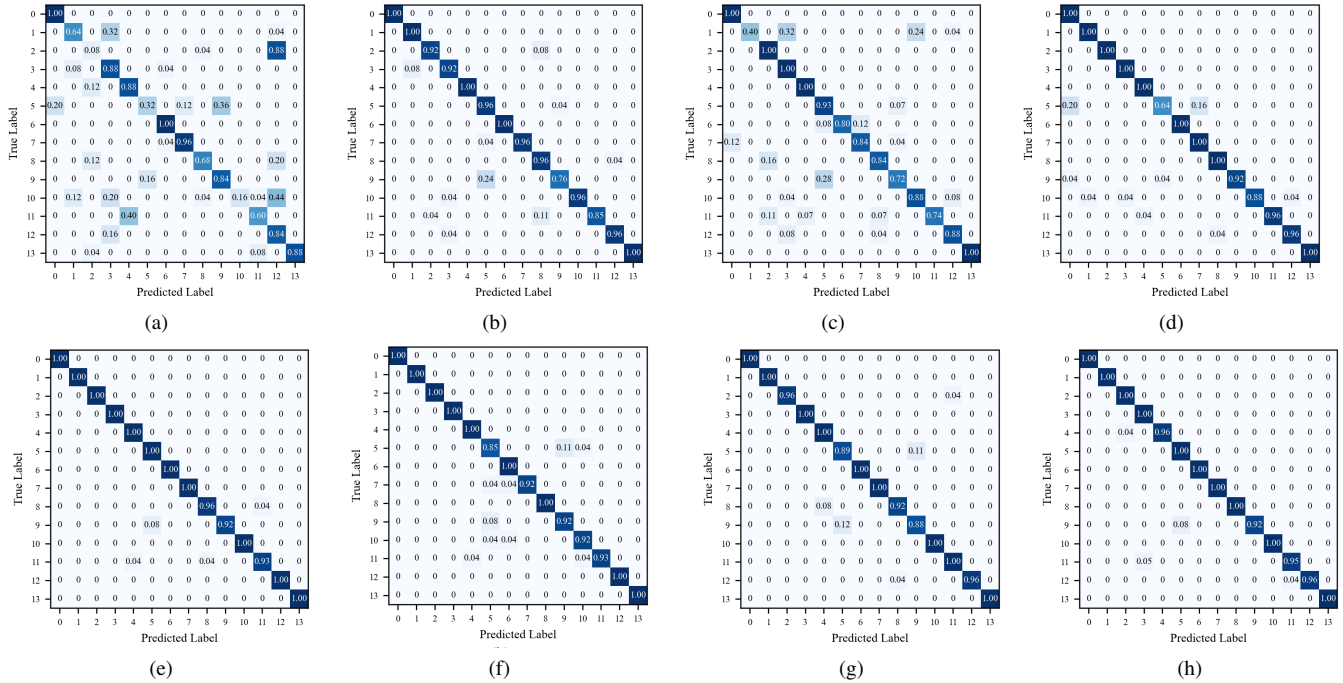


Fig. 6. The confusion matrices of different methods on the PU dataset in A1→A3. (a) MK-MMD; (b) DANN; (c) MoCo; (d) DACN; (e) MCDDFA; (f) TF-C; (g) CCLP; (h) MSTF-CN.

degradation patterns, where off-diagonal elements are more prominent, indicating their limited discriminative capability when operating conditions change. These results suggest that existing methods struggle to maintain discriminative capability when facing distribution shifts, whereas MSTF-CN achieves more accurate fault diagnosis.

To further evaluate the feature representation capability, t-distributed stochastic neighbor embedding (t-SNE) is employed to visualize the learned feature distributions, as shown in Fig. 8-9 for the PU and HUST datasets. Each point represents a sample, and different colors correspond to different fault categories. The results show that the features obtained by

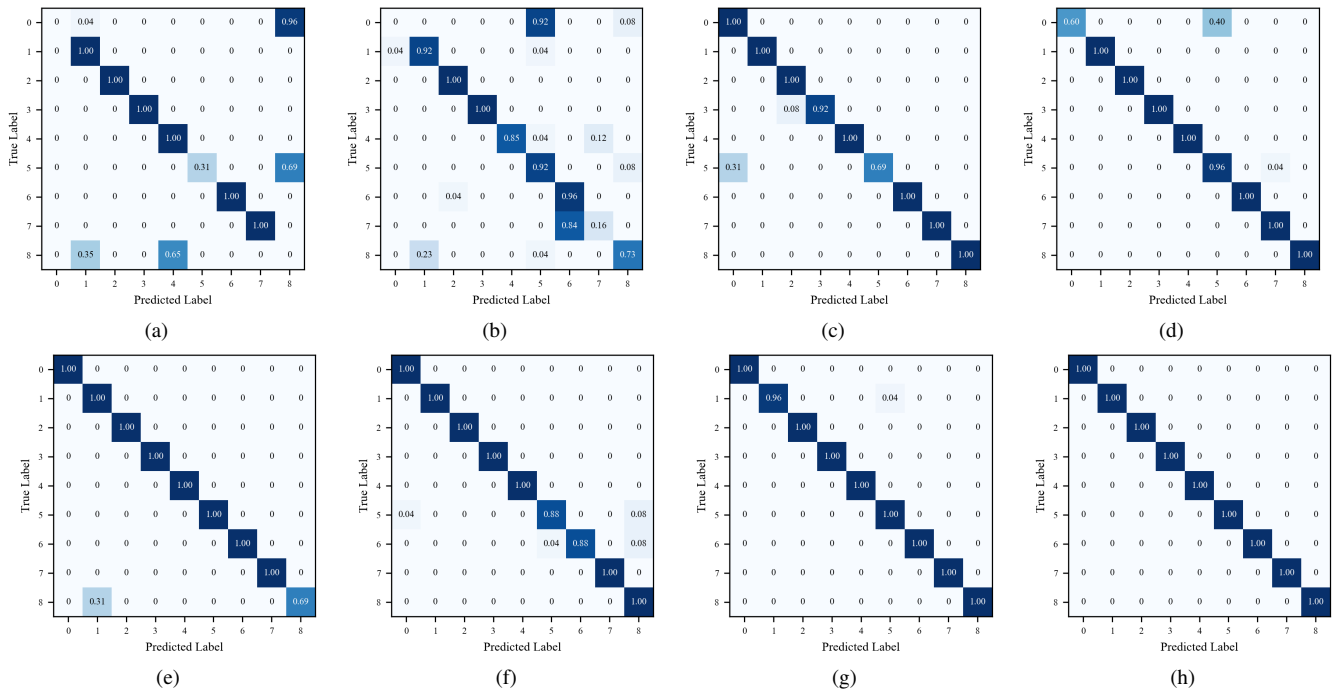


Fig. 7. The confusion matrices of different methods on the HUST dataset in B1→B3. (a) MK-MMD; (b) DANN; (c) MoCo; (d) DACN; (e) MCDDFA; (f) TF-C; (g) CCLP; (h) MSTF-CN.

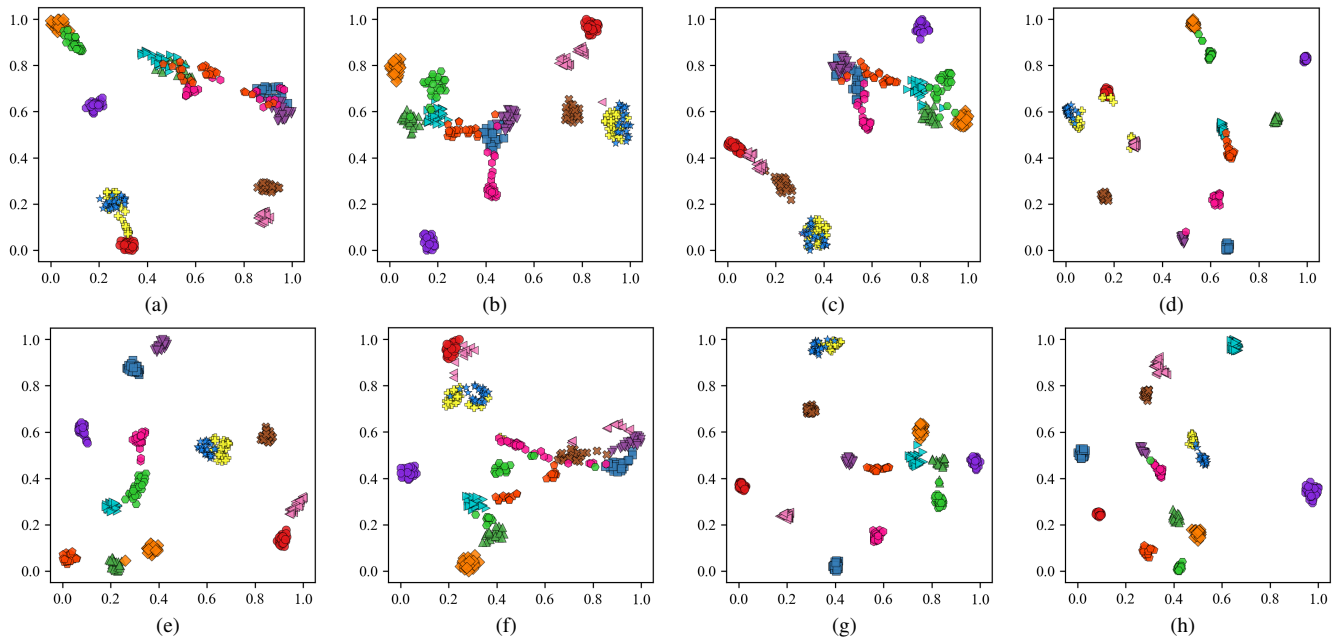


Fig. 8. The t-SNE visualizations of different methods on the PU dataset in A1→A3. (a) MK-MMD; (b) DANN; (c) MoCo; (d) DACN; (e) MCDDFA; (f) TF-C; (g) CCLP; (h) MSTF-CN.

MSTF-CN form more compact and well-separated clusters, with clearer decision boundaries between different classes. Samples belonging to the same category are tightly grouped, whereas clusters of different categories remain distinguishable even under varying operating conditions. In contrast, features learned by the baseline methods exhibit noticeable overlap across several classes, indicating insufficient feature separability and weak domain-invariant representation learning. These observations further confirm that the proposed method

can learn more discriminative feature representations, which directly contribute to its improved performance in the cross-domain fault diagnosis.

2) Analysis of Graph Construction Strategy: To further investigate the effect of the source-domain graph construction strategy on domain adaptation, we compare the proposed label-guided graph construction with a similarity-based top-k graph construction. As shown in Fig. 10, the proposed label-guided graph construction consistently achieves better performance

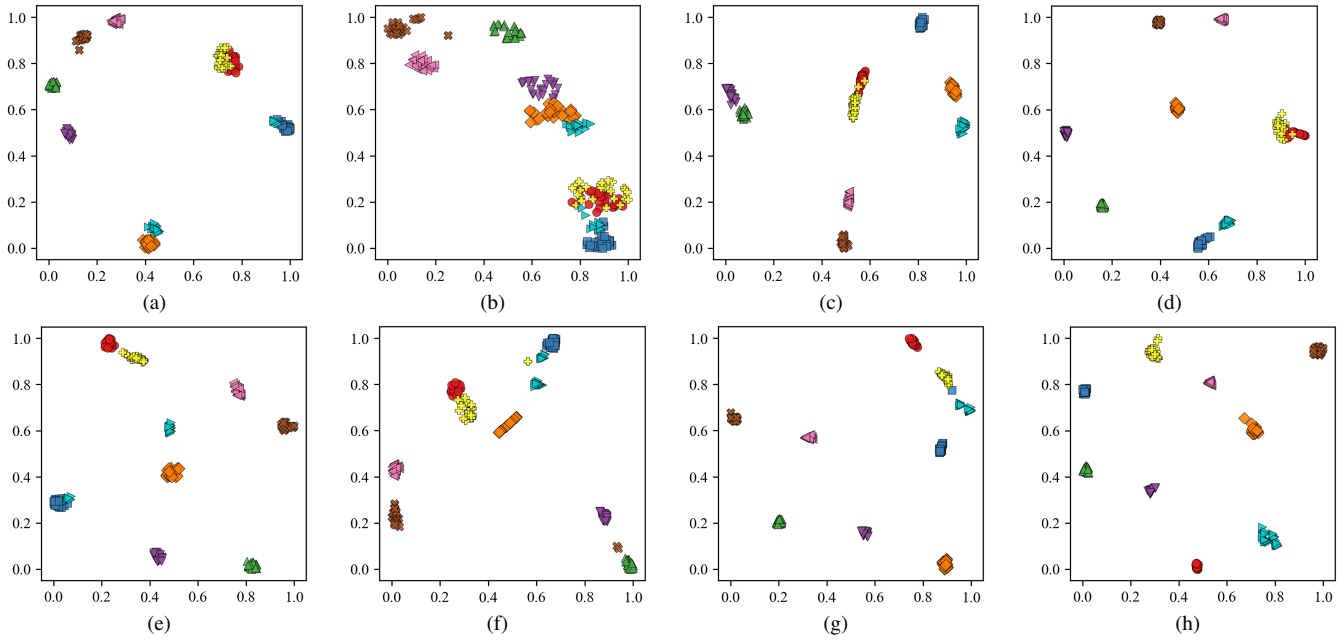


Fig. 9. The t-SNE visualizations of different methods on HUST dataset in B1→B3. (a) MK-MMD; (b) DANN; (c) MoCo; (d) DACN; (e) MCDDFA; (f) TF-C; (g) CCLP; (h) MSTF-CN.

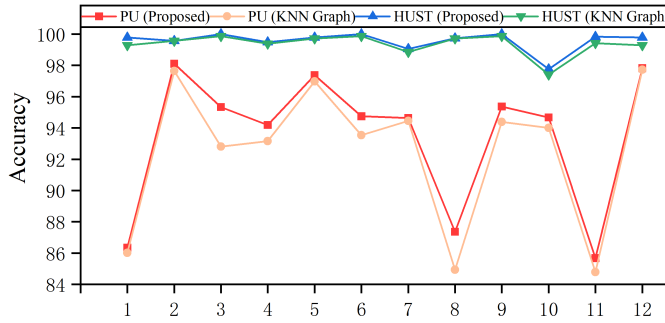


Fig. 10. Effect of graph construction strategy on accuracy over 12 transfer tasks.

across all transfer tasks on both the PU and HUST datasets. This is because the label-guided graph explicitly establishes semantic relationships among samples from the same fault category, enabling graph convolution to aggregate information from semantically consistent neighbors. In contrast, constructing the source-domain graph solely based on feature similarity cannot fully exploit the available supervisory information. These results demonstrate that utilizing source-domain labels during graph construction facilitates the learning of more discriminative graph representations, thereby improving cross-domain fault diagnosis performance.

3) Evaluation on the Influence of Parameters: The proposed MSTF-CN involves several important parameters, among which the neighborhood size k in graph construction and the contrastive loss weight λ play critical roles in structural representation learning and feature alignment. Therefore, a joint sensitivity analysis is conducted by evaluating different combinations of k and λ on the PU and HUST datasets. As shown in Fig. 11-12, the performance varies with different parameter combinations, and the optimal configuration is

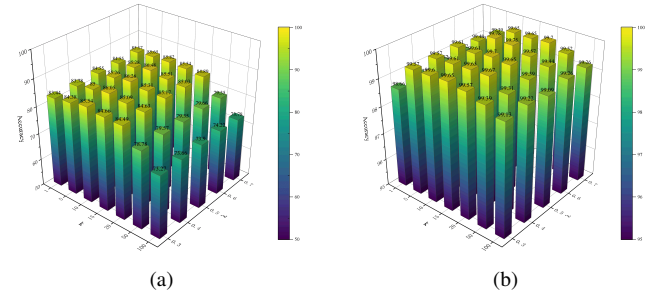


Fig. 11. Parameter sensitivity analysis under the A1→A2 transfer task on the PU dataset: (a) Accuracy; (b) F1-score.

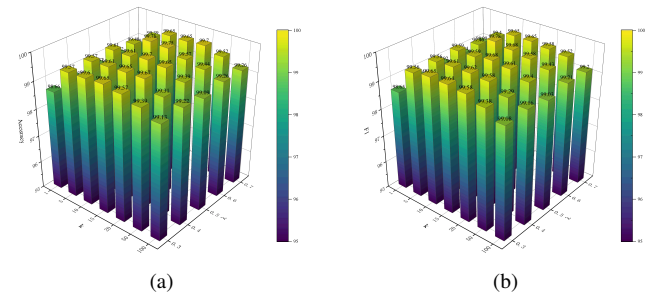


Fig. 12. Parameter sensitivity analysis under the B1→B2 transfer task on the HUST dataset: (a) Accuracy; (b) F1-score.

obtained at $k = 10$ and $\lambda = 0.6$. A relatively small k limits the aggregation of useful neighborhood information, while an excessively large k introduces irrelevant connections and weakens the discriminative graph structure. Meanwhile, an inappropriate value of λ may disturb the balance between classification supervision and contrastive representation alignment. These results indicate that the selected parameter combination provides an appropriate trade-off between structural informa-

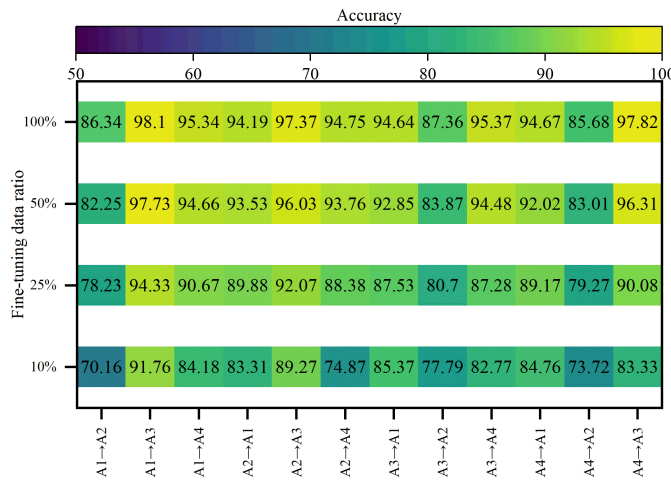


Fig. 13. Effect of fine-tuning data ratio on the PU datasets.

tion propagation and domain-invariant feature learning.

In addition, the influence of labeled target-domain data is further investigated to evaluate the adaptation capability of MSTF-CN under different annotation budgets. Specifically, the fine-tuning ratio is set to 10%, 25%, 50%, and 100% of the target-domain training samples on the PU dataset. The corresponding average accuracies are 81.77%, 87.30%, 91.71%, and 93.47%, respectively, as shown in Fig. 13. When only 10% of the target-domain training samples are utilized for fine-tuning, MSTF-CN still achieves an accuracy of 81.77%, where each fault category contains only approximately 2 labeled samples on average. This demonstrates that the proposed framework can effectively exploit unlabeled target-domain information through multi-sensor graph representation learning and cross-sensor contrastive alignment. Moreover, the performance improvement gradually decreases as the proportion of labeled target samples increases, indicating that MSTF-CN can achieve effective domain adaptation with limited target-domain annotations.

4) Ablation Study: To further evaluate the contribution of each component in MSTF-CN, ablation experiments were conducted on the PU and HUST datasets under 12 cross-domain transfer tasks. Five ablation variants were constructed by removing individual components, including “w/o GCN”, “w/o CSCA”, “w/o CSCA”, “w/o IRCL”, “w/o IECL”, and “w/o UFF”, where “w/o GCN” replaces the GCN encoder with MLP. As shown in Figs. 14-15, removing any component leads to performance degradation on most transfer tasks. First, replacing GCN with MLP results in a noticeable performance decrease on both datasets. Unlike MLP, which independently processes each sample, GCN effectively captures structural correlations through neighborhood information aggregation, leading to more discriminative graph representations. Removing the CSCA module leads to the most significant performance drop across both datasets. Without CSCA, the model degenerates into a representation learning scheme without explicit alignment constraints, causing features from different sensors and domains to remain scattered in the embedding space. As a result, representations corresponding to the same

TABLE V

EXPERIMENTAL RESULTS OF COMPLEXITY ANALYSIS ON THE PU DATASET

Methods	FLOPs (M)	Training time (s)	Inference time (ms)
MK-MMD	7.86	0.26	0.67
CORAL	7.86	0.19	0.70
DANN	27.22	0.40	1.22
MoCo	58.23	0.56	0.97
SimCLR	58.24	0.63	1.03
WSCN	14.49	0.77	1.55
DSCN	28.97	1.13	2.34
CA-DGNN	10.90	0.67	2.25
MCDDFA	41.97	0.89	0.84
TF-C	20.46	0.98	1.20
CCLP	76.91	1.05	1.92
MSTF-CN	13.77	0.63	1.01

fault state are no longer consistently clustered, leading to degraded transferability and unstable decision boundaries under domain shift. This observation confirms that explicit alignment is more critical than feature extraction alone in cross-domain multi-sensor scenarios. Similarly, removing IRCL or IECL also leads to noticeable performance degradation, demonstrating the effectiveness of both intra-sensor and inter-sensor contrastive alignment strategies. Without IRCL, the consistency between time- and frequency-domain representations within each sensor is reduced, making the learned features more sensitive to operating-condition variations. Without IECL, the alignment between different sensors is weakened, limiting the ability of the model to exploit complementary multi-sensor information. These results indicate that both intra-sensor and inter-sensor alignment contribute to constructing more stable and discriminative feature representations. Finally, removing the UFF module also leads to performance degradation, although the reduction is relatively limited. This indicates that the proposed UFF module mainly plays a complementary enhancement role by promoting the interaction between time and frequency features, thereby further improving feature complementarity and representation refinement.

5) Complexity Analysis: To evaluate the computational efficiency of the proposed MSTF-CN, the complexity of the model and the performance of the runtime are compared with several representative methods. In this experiment, the training time denotes the average time required per epoch, while the inference time represents the average time consumed to process a single sample. As shown in Table V, the proposed MSTF-CN achieves a well-balanced trade-off between computational cost and representation capability. From a computational perspective, MSTF-CN requires 13.77 M FLOPs, which is substantially lower than those of several deep learning-based approaches. In terms of computational efficiency, the training time of MSTF-CN remains at a moderate level and is comparable to that of mainstream deep learning models, while being lower than that of more complex methods. Overall, MSTF-CN maintains relatively low computational complexity and competitive runtime efficiency, while achieving superior diagnostic performance as demonstrated in previous sections. This confirms that the proposed method attains an effective balance between accuracy and efficiency, making it well-suited

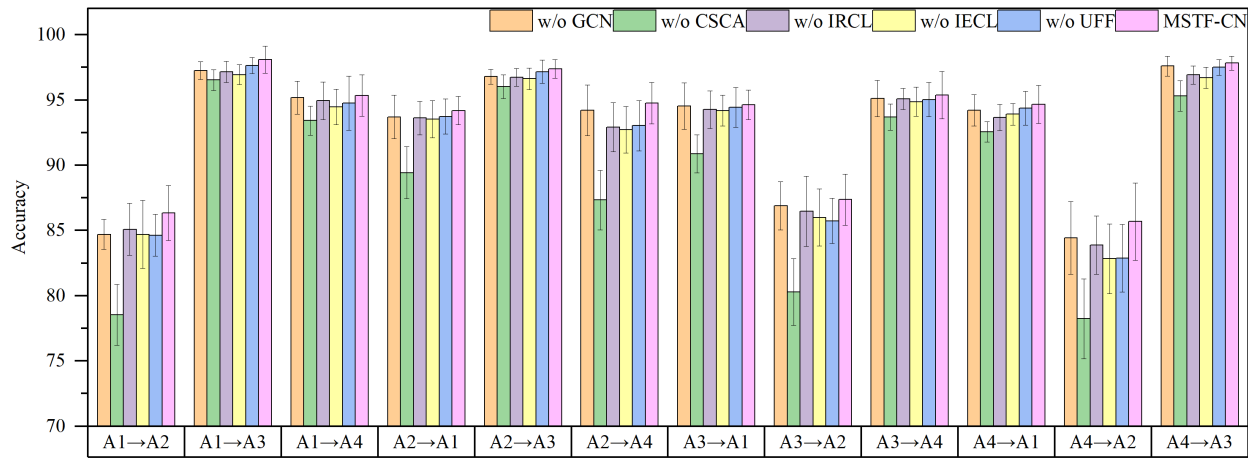


Fig. 14. Accuracy of ablation models under different transfer tasks on the PU dataset.

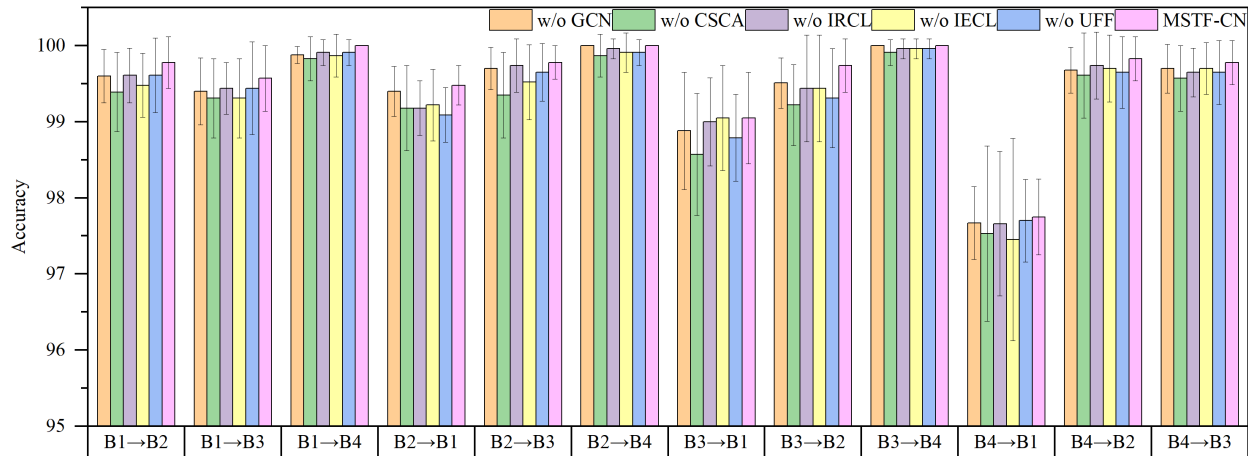


Fig. 15. Accuracy of ablation models under different transfer tasks on the HUST dataset.

for real-world multi-sensor cross-domain fault diagnosis tasks.

V. CONCLUSION

In this article, A novel DA fault diagnosis framework, termed MSTF-CN, is proposed to address performance degradation caused by distribution discrepancies under varying operating conditions. Specifically, a unified representation learning architecture is developed to capture discriminative features from multi-sensor vibration signals while alleviating the adverse effects of domain shifts. By incorporating collaborative learning in both time and frequency domains, the method enhances the expressiveness and completeness of the learned features, enabling a more effective extraction of fault-related information in cross-domain scenarios. In addition, the multi-sensor representation learning mechanism facilitates effective fusion of heterogeneous sensor information, thereby improving the reliability and stability of the learned representations. Extensive experiments conducted on two widely used bearing datasets demonstrate that the proposed method consistently outperforms several representative SOTA approaches in multiple transfer tasks. Nevertheless, the current framework still

partially relies on labeled samples in the target domain during the adaptation stage, which may limit its scalability in more challenging fully unsupervised scenarios. Future work will focus on exploring more flexible learning paradigms, such as continual learning [41] and open-set [42] fault diagnosis, to further enhance the generalization capability and practical applicability of the proposed framework.

REFERENCES

- [1] J. Luo et al., "A multimodal IoT sensing system with hybrid LSTM-CNN model for predictive fault diagnosis in elevators," *IEEE Sensors J.*, vol. 25, no. 20, pp. 38829–38836, Oct. 2025.
- [2] L. Liang, S. Liu, M. Zhong, and R. Su, "Distributed fault detection for cyber-physical systems with application to power network system," *IEEE Trans. Cybern.*, vol. 55, no. 11, pp. 5117–5126, Nov. 2025.
- [3] Y. Li, J. Yang, W. Wang, and T. Gao, "A joint collaborative adaptation network for fault diagnosis of rolling bearing under class imbalance and variable operating conditions," *Adv. Eng. Informat.*, vol. 69, Jan. 2026, Art. no. 103931.
- [4] C. Fan, Y. Zhang, H. Ma, K. Yu, and Z. Ma, "A novel lightweight DDPM-based data augmentation method for rotating machinery fault diagnosis with small sample," *Mech. Syst. Signal Process.*, vol. 232, Jun. 2025, Art. no. 112741.

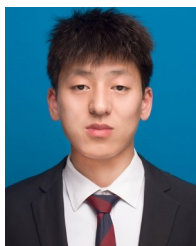
- [5] D. Zhou, Q. Yao, H. Wu, S. Ma, and H. Zhang, "Fault diagnosis of gas turbine based on partly interpretable convolutional neural networks," *Energy*, vol. 200, Jun. 2020, Art. no. 117467.
- [6] M. Mansouri, K. Dhibi, M. Hajji, K. Bouzara, H. Nounou, and M. Nounou, "Interval-valued reduced RNN for fault detection and diagnosis for wind energy conversion systems," *IEEE Sensors J.*, vol. 22, no. 13, pp. 13581–13588, Jul. 2022.
- [7] H. Liu, Q. Xu, X. Han, B. Wang, and X. Yi, "Attention on the key modes: Machinery fault diagnosis transformers through variational mode decomposition," *Knowl.-Based Syst.*, vol. 289, Apr. 2024, Art. no. 111479.
- [8] X. Yan, W.-J. Yan, Y. Xu, and K.-V. Yuen, "Machinery multi-sensor fault diagnosis based on adaptive multivariate feature mode decomposition and multi-attention fusion residual convolutional neural network," *Mech. Syst. Signal Process.*, vol. 202, Nov. 2023, Art. no. 110664.
- [9] X. Zhang, X. Zhang, J. Liu, B. Wu, and Y. Hu, "Graph features dynamic fusion learning driven by multi-head attention for large rotating machinery fault diagnosis with multi-sensor data," *Eng. Appl. Artif. Intell.*, vol. 125, Oct. 2023, Art. no. 106601.
- [10] C. Yang, J. Liu, K. Zhou, X. Yuan, and X. Jiang, "A meta-path graph-based graph homogenization framework for machine fault diagnosis," *Eng. Appl. Artif. Intell.*, vol. 121, May 2023, Art. no. 105960.
- [11] X. Xiao, C. Li, H. He, J. Huang, and T. Yu, "Rotating machinery fault diagnosis method based on multi-level fusion framework of multi-sensor information," *Inf. Fusion*, vol. 113, Jan. 2025, Art. no. 102621.
- [12] J. Jiao, M. Zhao, J. Lin, and C. Ding, "Classifier inconsistency-based domain adaptation network for partial transfer intelligent diagnosis," *IEEE Trans. Ind. Inform.*, vol. 16, no. 9, pp. 5965–5974, Sep. 2020.
- [13] D. J. Sutherland et al., "Generative models and model criticism via optimized maximum mean discrepancy," *arXiv:1611.04488*, 2016.
- [14] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, 2016.
- [15] W. Zellinger, T. Grubinger, E. Lughofer, T. Natschlager, and S. Saminger-Platz, "Central moment discrepancy (CMD) for domain-invariant representation learning," *arXiv:1702.08811*, 2017.
- [16] X. Cao, B. Chen, and N. Zeng, "A deep domain adaption model with multi-task networks for planetary gearbox fault diagnosis," *Neurocomputing*, vol. 409, pp. 173–190, Oct. 2020.
- [17] B. Wang, B. Wang, and Y. Ning, "A novel transfer learning fault diagnosis method for rolling bearing based on feature correlation matching," *Meas. Sci. Technol.*, vol. 33, no. 12, Dec. 2022, Art. no. 125006.
- [18] J. Si, H. Shi, J. Chen, and C. Zheng, "Unsupervised deep transfer learning with moment matching: A new intelligent fault diagnosis approach for bearings," *Measurement*, vol. 172, Feb. 2021.
- [19] K. Zhao, H. Jiang, K. Wang, and Z. Pei, "Joint distribution adaptation network with adversarial learning for rolling bearing fault diagnosis," *Knowl.-Based Syst.*, vol. 222, Jun. 2021, Art. no. 106974.
- [20] M. Deng, A. Deng, Y. Shi, and M. Xu, "Correlation regularized conditional adversarial adaptation for multi-target-domain fault diagnosis," *IEEE Trans. Ind. Inform.*, vol. 18, no. 12, pp. 8692–8702, Dec. 2022.
- [21] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3733–3742.
- [22] Y. Zhou, H. Wang, G. Wang, A. Kumar, W. Sun, and J. Xiang, "Semi-supervised multiscale permutation entropy-enhanced contrastive learning for fault diagnosis of rotating machinery," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 3525610.
- [23] Y. An, K. Zhang, Y. Chai, Q. Liu, and X. Huang, "Bearing fault diagnosis under variable working conditions base on contrastive domain adaptation method," *IEEE Trans. Instrum. Meas.*, vol. 71, 2022, Art. no. 3522711.
- [24] Y. Pang, Q. Liu, Z. Sun, Z. Xu, and Z. Hao, "Time-frequency supervised contrastive learning via pseudo-labeling: An unsupervised domain adaptation network for rolling bearing fault diagnosis under time-varying speeds," *Adv. Eng. Informat.*, vol. 59, Jan. 2024, Art. no. 102304.
- [25] Y. Zhang, J. Hua, D. Zhang, J. He, and X. Fang, "train bearing fault diagnosis based on time-frequency signal contrastive domain share CNN," *IEEE Sensors J.*, vol. 24, no. 20, pp. 33669–33681, Oct. 2024.
- [26] L. Wang, Y. Gao, L. Gao, and X. Li, "Adversarial contrastive learning-enabled domain extension method for time-varying cross-domain fault diagnosis," *IEEE Trans. Instrum. Meas.*, vol. 74, 2025, Art. no. 3552713.
- [27] X. Cao, L. Zhao, Y. Xia, and H. Yang, "Sensor fusion with feedback for robust target tracking in autonomous ground vehicles under missing measurements," *IEEE Trans. Veh. Technol.*, vol. 75, no. 2, pp. 1854–1865, Feb. 2026.
- [28] Y. He, C. Zhao, and W. Shen, "Cross-domain compound fault diagnosis of machine-level motors via time-frequency self-contrastive learning," *IEEE Trans. Ind. Inform.*, vol. 20, no. 7, pp. 9692–9701, Jul. 2024.
- [29] C. Lessmeier, J. K. Kimotho, D. Zimmer, and W. Sextro, "Condition monitoring of bearing damage in electromechanical drive systems by using motor current signals of electric motors: A benchmark data set for data-driven classification," in *Proc. PHM Soc. Eur. Conf.*, 2016.
- [30] C. Zhao, E. Zio, and W. Shen, "Domain generalization for cross-domain fault diagnosis: An application-oriented perspective and a benchmark study," *Reliab. Eng. Syst. Saf.*, vol. 245, May 2024, Art. no. 109964.
- [31] M. Long, Y. Cao, J. Wang, and M. Jordan, "Learning transferable features with deep adaptation networks," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 97–105.
- [32] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.
- [33] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9729–9738.
- [34] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
- [35] R. Huang, J. Li, S. Wang, G. Li, and W. Li, "A Robust Weight-Shared Capsule Network for Intelligent Machinery Fault Diagnosis," *IEEE Trans. Ind. Inform.*, vol. 16, no. 10, pp. 6466–6475, Oct. 2020.
- [36] R. Huang, J. Li, Y. Liao, J. Chen, Z. Wang, and W. Li, "Deep Adversarial Capsule Network for Compound Fault Diagnosis of Machinery Toward Multidomain Generalization Task," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.
- [37] R. Wang, Z. Lyu, F. Yan, L. Yu, and X. Hu, "A conditional adaption alignment dynamic graph neural network model for unsupervised fault diagnosis of rotating machinery," *Mech. Syst. Signal Process.*, vol. 240, Nov. 2025, Art. no. 113361.
- [38] J. Li, W. Deng, X. Dang, and H. Zhao, "Cross-domain adaptation fault diagnosis with maximum classifier discrepancy and deep feature alignment under variable working conditions," *IEEE Trans. Reliab.*, vol. 74, no. 3, pp. 4106–4115, Sep. 2025.
- [39] X. Zhang, Z. Zhao, T. Tsiligkaridis, and M. Zitnik, "Self-supervised contrastive pre-training for time series via time-frequency consistency," *Adv. Neural Inf. Process. Syst.*, vol. 35, pp. 3988–4003, 2022.
- [40] H. Hu et al., "Cross-sensor contrastive learning-based pre-training for machinery fault diagnosis under sample-limited conditions," *Knowl.-Based Syst.*, vol. 311, Feb. 2025, Art. no. 113075.
- [41] B. Chen, X. Zhang, C. Shen, Q. Li, and Z. Song, "CoUDA: Continual Unsupervised Domain Adaptation for Industrial Fault Diagnosis Under Dynamic Working Conditions," *IEEE Trans. Ind. Inform.*, vol. 21, no. 5, pp. 4072–4082, May 2025.
- [42] G. Huang et al., "A noncontact open-set fault diagnosis method based on latent space disentanglement and prototype representation," *IEEE Sensors J.*, vol. 26, no. 4, pp. 6506–6518, Feb. 2026.



Gang Wang (Member, IEEE) received the Ph.D. degree in management science and engineering from the School of Management, Fudan University, Shanghai, China, in 2008.

He is currently a Professor with the School of Management, Hefei University of Technology, Hefei, China. His past research has been published in *Management Information Systems Quarterly*, *IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT*, *IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS*, *IEEE INTELLIGENT SYSTEMS*, *Decision Support Systems* and in top-tier conferences, including the International Conference on Information Systems, the Pacific Asia Conference on Information Systems, and the Hawaii International Conference on System Sciences. His current research interests include industrial big data analytics, business analytics, Fintech, and machine learning.

Dr. Wang serves as an Associate Editor for *Decision Support Systems*.



Zhihui Yao received the B.S. degree from the Hefei University of Technology, Hefei, China, in 2024. He is currently pursuing the M.S. degree with the School of Management, Hefei University of Technology, Hefei, China.

His research interests include intelligent manufacturing and fault diagnosis.



Can Zhang received the B.S. degree from the Jiangxi University of Finance and Economics, Nanchang, China, in 2024. She is currently pursuing the M.S. degree with the School of Management, Hefei University of Technology, Hefei, China.

Her research interests include intelligent manufacturing and fault diagnosis.



Xiaohui Yuan (Senior Member, IEEE) received the B.S. degree in electrical engineering from the Hefei University of Technology, Hefei, China, in 1996, and the Ph.D. degree in computer science from Tulane University, New Orleans, LA, USA, in 2004. He is currently a tenured Associate Professor with the Department of Computer Science and Engineering, University of North Texas, Denton, TX, USA.

His research interests include computer vision, machine learning, and artificial intelligence.



Mingfeng Lu (Student Member, IEEE) received the B.S. degree from the Hefei University of Technology, Hefei, China, in 2020, where she is currently pursuing the Ph.D. degree in management science and engineering.

Her research interests include deep learning, intelligent manufacturing, and fault diagnosis and prognosis.